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Data Structure lab 5

Task 1 :

```
#include<iostream>
#include<stack>
using namespace std;
//  () {} []

bool isBalanced(string str) {
    stack<char>s;

    //parsing through stack
    for (int i = 0; i < str.length(); ++i) {

        if (str[i] != '(' || str[i] != ')' || str[i] != '{' || str[i] != '}' ||
str[i] != '[' || str[i] != ']') {
            continue;
        }
        //opening pushed to stack
        if (str[i] == '(' || str[i] == '{' || str[i] == '[')
            s.push(str[i]);

        else {

            if (!s.empty()) {

                //getting top element to check
                char t = s.top();

                //if top == character then popping top
                if (str[i] == ')' && t == '('
                    || str[i] == '}' && t == '{'
                    || str[i] == ']' && t == '[')
                {
                    s.pop();
                }
            }
        }
    }
}
```

```

        //if condition not met then false
        else {
            return false;
        }
    }
    else {
        return false;
    }
}

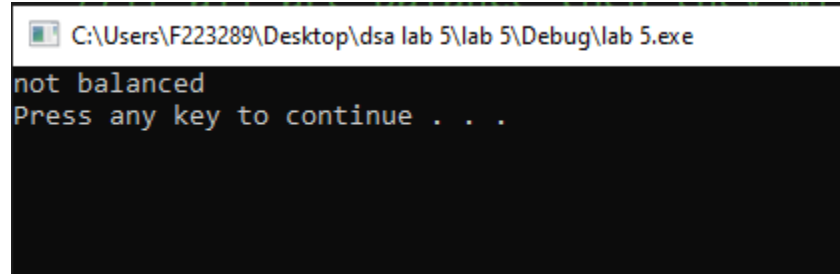
//if all are balance then they will pop and stack will be empty in end
elsewise not balanced
if (s.empty())
    return true;
else
    return false;
}

int main()
{
    string str = "{}()[]";
    if (isBalanced(str))
    {
        cout << "balanced \n";
    }
    else
        cout << "not balanced \n";

    system("pause");
    return 0;
}

```

Output:



```

C:\Users\F223289\Desktop\dsa lab 5\lab 5\Debug\lab 5.exe
not balanced
Press any key to continue . . .

```

Task 2:

```

#include<iostream>
#include<stack>

```

```

using namespace std;
bool isPalindrome(string str) {

    stack<char>s;

    // pushing whole string to stack
    for (int i = 0; i < str.length(); ++i) {
        s.push(str[i]);
    }

    //stack will have ending characters and string will have starting so match and pop
    //if not match then not palindrome

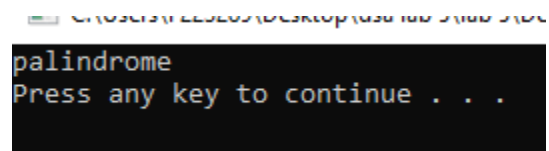
    for (int i = 0; i < (s.size() / 2); ++i) {
        if (str[i] == s.top()) {
            s.pop();
            continue;
        }
        else {
            return false;
        }
    }

    return true;
}

int main()
{
    string str = "alla";
    if (isPalindrome(str)) {
        cout << "palindrome \n";
    }
    else {
        cout << "not palindorme \n";
    }

    system("pause");
    return 0;
}

```



```

C:\Users\LEO\Desktop\code\100\100\100\100
palindrome
Press any key to continue . . .

```

Task 3 :

```
#include <iostream>
#include<string>
#include<stack>
using namespace std;
int prec(char c)
{
    if (c == '^')
        return 3;
    else if (c == '/' || c == '*')
        return 2;
    else if (c == '+' || c == '-')
        return 1;
    else
        return -1;
}
void infixToPostfix(string s)
{
    stack<char> st;
    string result;

    for (int i = 0; i < s.length(); i++) {
        char c = s[i];

        if ((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z')
            || (c >= '0' && c <= '9'))
            result += c;

        else if (c == '(')
            st.push('(');

        else if (c == ')') {
            while (st.top() != '(') {
                result += st.top();
                st.pop();
            }
            st.pop();
        }

        else {
            while (!st.empty()
                && prec(s[i]) <= prec(st.top())) {
                result += st.top();
                st.pop();
            }
            st.push(c);
        }
    }

    while (!st.empty()) {
        result += st.top();
        st.pop();
    }
}
```

```
        cout << result << endl;
    }

    int main()
    {
        string exp = "a+b*(c^d-e);

        infixToPostfix(exp);

        return 0;
    }
```